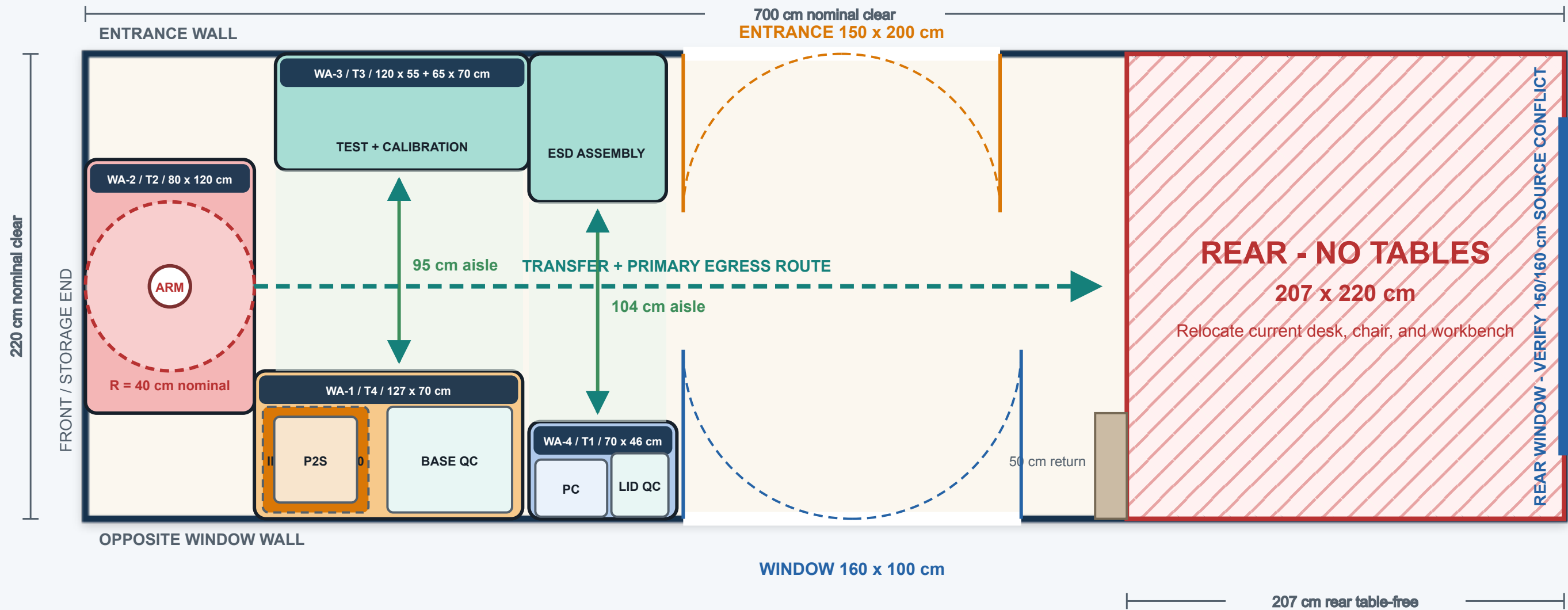
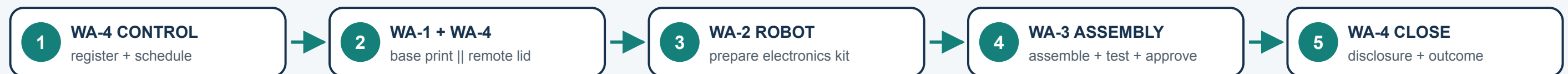


Nominal clear interior: 7.00 x 2.20 m

Front/storage end at left; rear window end at right. Entrance is on the upper long wall; the 160 x 100 cm window is opposite.

**Bayreuth distributed-route hand-offs****Design basis**

Drawing-derived interior 700 x 220 cm; entrance 150 x 200 cm; opposite window 160 x 100 cm; rear bay 207 cm. Nominal aisle: 95 cm. ASR A2.3 concept target: at least 90 cm for up to five people; confirm applicability on site.

Installation hold points

Survey table load/height, opening coordinates, outlets, cable paths, P2S ventilation, and rear furniture removal.

PAROL6 guarding, mounting, E-stop, and safe reset require an independent risk assessment. Printed fixtures are not safety devices.

Design basis

A compact, forward-loaded four-station cell that protects the complete right/rear bay from table encroachment and keeps the process route legible.

7.00 × 2.20 m nominal clear interior	4 documented table footprints	2.07 × 2.20 m rear table-free zone	95 cm nominal opposing aisle
Why this arrangement Design intent. WA-2 uses the 80 × 120 cm table at the front for the robot kit cell; WA-1 uses the 127 × 70 cm table on the opposite wall for printing and base inspection; WA-3 uses the stepped table to separate assembly from test; WA-4 uses the compact 70 × 46 cm table beside the entrance for control and receiving. The mapping is a documented design inference because the source table drawing names dimensions but does not bind a table to a work area.		How to read readiness Evidence posture. Workbook quantities mean <i>listed</i>, not installed or operational. Only specific infrastructure supported by repository documentation or photographs is marked verified-existing. Every listed, assigned, configured, acquired, or verified item still requires a physical commissioning record.	

Dimensioned table assignment

Area / table	Footprint	Room coordinates	Placement rationale
WA-1 · T4 Print / base quality	127 x 70 cm	x 80–207 cm; y 0–70 cm	The 70 cm depth and 127 cm length hold the P2S installation envelope and a separate base-inspection zone.
WA-2 · T2 Kit preparation	80 x 120 cm	x 0–80 cm; y 50–170 cm	End-cell orientation contains the documented 40 cm nominal PAROL6 reach over the tabletop; the certified guard envelope remains a hold point.
WA-3 · T3 Assembly / test	120 x 55 + 65 x 70 cm	x 90–275 cm; y 150–220 cm	The stepped surface separates the deep ESD assembly area from the long test/calibration area while preserving the aisle.
WA-4 · T1 Control / lid	70 x 46 cm	x 210–280 cm; y 0–46 cm	The compact multi-visit station is nearest the entrance and directly opposite the assembly hand-off.

Aisle hold point. The calculated 95 cm opposing aisle is compared with a 90 cm planning target from BAuA ASR A2.3 for the stated small-occupancy case. This is a concept check, not a compliance finding; door operation, protrusions, work postures and the measured escape route still require competent evaluation. Source: official BAuA ASR A2.3 PDF; full link is recorded in the primary-reference register.

Survey hold points before furniture is moved

Check	Required evidence	Why it matters
Clear shell and openings	Laser-measured wall-to-wall dimensions, entrance/window positions, door swing, sill and handle envelopes	The concept uses the 7.00 × 2.20 m nominal clear dimensions derived from the source floorplan.
Tables	Top height, exact footprint, wheel/brake state, load rating, deflection and edge profile	Robot mounting, printer service access and the printed cable fixtures depend on these values.
Utilities and fixed equipment	Outlet/circuit map, network ports, fire equipment, first aid, HVAC, rack and display clearances	The rear bay remains table-free, but fixed services and access routes must also remain usable.
Rear short-wall window	Measured opening width and operating envelope	The scan indicates 160 cm while the existing OBJ geometry indicates 150 cm; neither is silently preferred for construction.

CONCEPT ONLY — measured survey, load verification, fire/electrical review, and machinery/robot risk assessment are required before installation or use.

WA-1 · Print / base quality

T4: Rectangular table, 127 x 70 cm. Placement x 80–207 cm; y 0–70 cm.

Process fragment Print base · Inspect base · Review base · Schedule reprint energy · Release base	Physical flow Input: print job, material Output: accepted base	Evidence package authorization ID · print-job ID · component ID · measured energy · printer result · inspection decision · tool-slot state and sensor health · evidence timestamps
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Hardware at the table

Status	Hardware / quantity	Function at station	Commissioning action
VERIFY	P2S 3D printer and local telemetry 1 unit	Print the enclosure base and provide authoritative job lifecycle data.	Verify the exact model, API, serial number, operational state, service clearance, ventilation, and fire controls before assignment.
ACQUIRE	Calibrated plug-level energy meter 1 unit	Measure printer power and cumulative energy for each print job.	No dedicated plug-level energy meter is listed in the workbook.
VERIFY	MLX90640 thermal array 1 unit	Observe printer heating and cooling without replacing machine interlocks.	Assign one of the two listed arrays only after an operational check.
VERIFY	XIAO ESP32S3 Sense or IMX335 workpiece camera 1 unit	Observe base presence and removal in a workpiece-only camera region.	Approve the privacy mask and retention policy before capture.
VERIFY	Keystudio Hall Magnetic tool-presence sensor and Nano input 1 monitored slot	Observe whether the designated process-tool slot is present, absent, or unknown without controlling any machine or safety function.	Verify the physical module, magnet tag, 5 V and logic compatibility, slot mapping, debounce, disconnect and stuck-state behavior. Slot absence is not evidence of tool use, correct work, or operator identity. Seven-slot coverage needs seven modules and verified additional input capacity; only five complete kits are inventory-listed.
CONFIGURE	Inspection button or user interface 1 station interface	Record base-quality, release, and reprint decisions explicitly.	A physical button or tablet UI may be selected during commissioning.
ASSIGN	Raspberry Pi edge-node package 1 package	Host printer and sensor adapters, timestamps, MQTT, and local buffering.	Assign one physically identified Pi and record its power and network path.

Printed fixtures assigned to this work area

Fixture	Qty	Envelope (mm)	Material / use	Fit gate
WA-1 cooled-part inspection tray	1	220 × 160 × 16	PETG. Keeps the printed base inside a repeatable camera and load-cell inspection region.	Usable inspection region is 212 x 152 mm. Verify the produced base envelope before printing.
WA-1 sensor-ready scraper and caliper rack	1	180 × 70 × 68	PETG. Separates hand tools from the cooled-part inspection surface and holds one removable Hall-sensor module behind the designated monitored slot.	Seven 10 mm open tool slots are provided. Tool handles wider than 10 mm rest in the front channel. The integrated cassette monitors one designated tool slot only. The official module envelope is 30 x 20 mm; PCB thickness, Hall-element position, connector keep-out, and component height require a physical fit check. Validate magnet distance and every allowed resting orientation; an empty slot is not evidence of correct tool use or operator identity. The presence sensor is research telemetry, not a safety interlock, and must not control the printer or protective functions. Monitoring all seven slots requires seven sensors and verified additional digital-input capacity; the workbook lists five complete sensor kits.
WA-1 optical and thermal sensor bridge bracket	1	90 × 55 × 71	PETG. Provides a non-safety adapter surface for a camera and MLX90640 sensor pod.	Optical opening is 44 x 28 mm. Hole spacing is a prototype and must be checked against the selected sensor carrier.

Printer allocation is a footprint concept, not a fire, fume, service-clearance, electrical-load, or unattended-operation approval. The Hall sensor observes one slot only; an empty slot does not prove tool use.

WA-2 · Kit preparation

T2: Rectangular table, 80 x 120 cm. Placement x 0–80 cm; y 50–170 cm.

Process fragment Robot kit	Physical flow Input: electronics bins Output: prepared kit	Evidence package robot-job ID · controller lifecycle · fault state · expected bin deltas · kit identity · workpiece-presence evidence
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Hardware at the table

Status	Hardware / quantity	Function at station	Commissioning action
VERIFY	PAROL6 robot arm and controller 1 system	Prepare the electronics kit and expose robot-job lifecycle data.	Verify controller, base fasteners, table load, reach envelope, serial connection, homing, and fault behavior before powered operation.
CONFIGURE	Instrumented component-bin load-cell set 1 set	Detect expected component removals and prepared-kit completion.	The final bin count follows the approved electronics bill of materials.
VERIFY	IMX335 or XIAO workpiece camera 1 unit	Observe component-bin, pickup, and prepared-kit presence.	Mount outside the robot reach and exclude the operator area.
ACQUIRE	Independent robot guard and emergency-stop system 1 system	Provide an independent safety boundary for powered robot motion.	Use certified safety hardware. Research sensing receives read-only state and must never implement or override the safety function.
CONFIGURE	PAROL6 controller event adapter 1 adapter	Bind controller lifecycle and fault records to the robot-job identity.	Validate disconnect, fault, stop, and recovery behavior.
ASSIGN	Raspberrry Pi edge-node package 1 package	Host controller and sensor adapters, synchronized time, and local buffering.	Install outside the robot reach and guarded operating envelope.

Printed fixtures assigned to this work area

Fixture	Qty	Envelope (mm)	Material / use	Fit gate
WA-2 instrumented component bin	4	110 × 85 × 55	PETG. Holds one identified component type over a dedicated weighing point.	Internal envelope is approximately 104 x 79 x 52 mm. Do not use for loose conductive parts without an ESD-safe liner.
WA-2 component-bin locator plate	4	122 × 97 × 8	PETG. Returns each component bin to the same load-cell and camera position.	Recess accepts a 110 x 85 mm bin base. Verify screw positions against the load-cell platform before drilling.
WA-2 prepared-kit tray	1	220 × 150 × 18	PETG with removable ESD-safe liner. Separates prepared electronics into three observable kit compartments.	Three equal nominal compartments are provided. The printed polymer is not assumed to be electrically dissipative.

Robot reach shown on the plan is geometric context only. A competent risk assessment must specify the guard, interlocks, emergency stop, mount, payload, speed, table rating, and access before powered motion.

WA-3 · Assembly / test

T3: L-shaped table: 65 x 70 cm leg plus 120 x 55 cm wing; overall 185 x 70 cm. Placement x 90–275 cm; y 150–220 cm.

Process fragment Assemble node · Test and calibrate · Resolve failure · Rework sensor · Approve deployment	Physical flow Input: base, lid, kit Output: approved sensor node	Evidence package product and component links · assembly start/end · test record · calibration record · rework history · final weight · approval
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Hardware at the table

Status	Hardware / quantity	Function at station	Commissioning action
CONFIGURE	Thin-film pressure mat 1 mat	Observe assembly start and component placement transitions.	Document sensor count, zoning, calibration, and strain relief in the mat design.
VERIFY	Workpiece camera 1 unit	Observe assembly state while excluding faces and adjacent stations.	Approve a workpiece-only region and retention policy before use.
CONFIGURE	Arduino Nano serial test jig 1 jig	Execute the versioned sensor-node test and calibration procedure.	Build and validate the jig; do not substitute the unfinished RX/TX path.
VERIFY	Reference scale and calibration references 1 set	Record completed-node weight and support versioned calibration checks.	Do not assume the WA-4 network scale can serve both stations. Acquire a second scale if no dedicated WA-3 unit is physically verified.
CONFIGURE	Approval button or user interface 1 station interface	Record explicit deployment approval after a passing test record.	Approval must remain an explicit human action.
ASSIGN	Raspberry Pi edge-node package 1 package	Host the serial receiver, evidence fusion, timestamps, and local buffer.	The documented Pi serial receiver remains incomplete and must be commissioned.

Printed fixtures assigned to this work area

Fixture	Qty	Envelope (mm)	Material / use	Fit gate
WA-3 base, lid, and kit staging tray	1	220 × 150 × 16	Validated ESD-safe PETG or PETG outside the ESD mat. Provides three separated identity-preserving staging pockets before assembly.	Each pocket is approximately 65 x 138 mm. Verify component envelopes and surface-resistivity requirements.
WA-3 Nano serial test-jig enclosure	1	110 × 80 × 28	Validated ESD-safe PETG. Protects a low-voltage carrier board and provides controlled USB cable exit.	Prototype post pattern is 58 x 36 mm. Carrier-board dimensions and electrical clearances require a physical fit check.
WA-3 reference-weight caddy	1	180 × 70 × 18	PETG. Keeps four calibration references identified and separated from active test parts.	Blind pockets have radii 18, 15, 12, and 9 mm. Measure the certified references before production printing.

Use validated ESD controls and certified calibration references. Printed polymer is not assumed dissipative without material evidence.

WA-4 · Control / receiving

T1: Rectangular table, 70 x 46 cm. Placement x 210–280 cm; y 0–46 cm.

Process fragment Register run · Select topology · Read solar and battery · Optimize energy-aware schedule · Select standard container · Select large container · Request matching lid · Evaluate decision · Set local fallback · Receive lid · Verify lid · Accept delivery · Request replacement · Apply disclosure · Share outcome	Physical flow Input: Munich lid, energy state Output: verified lid, process record	Evidence package run and work-order IDs · energy snapshot · schedule and policy versions · lid identity · delivery decision · disclosure decision · outcome
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Hardware at the table

Status	Hardware / quantity	Function at station	Commissioning action
VERIFIED EXISTING	Management PC 1 system	Operate run registration, scheduling, delivery, and outcome logging.	Retain network access while relocating the workstation to the selected table.
ACQUIRE	Read-only inverter and BMS adapter 1 adapter	Provide fresh, unit-qualified photovoltaic and battery snapshots.	Validate units, timestamps, freshness, disconnect behavior, and read-only access.
VERIFY	Network scale 1 unit	Observe lid receipt and removal after tare.	Verify reachability, protocol, credentials, tare, units, and calibration.
VERIFY	XIAO workpiece camera 1 unit	Observe lid presence and size cues in the receiving region.	Use identity capture as primary evidence; vision only corroborates it.
ACQUIRE	Two-dimensional identifier scanner 1 unit	Capture shipment, lid, size, and design-revision identities.	Provide validated manual entry only as the documented fallback.
CONFIGURE	MQTT broker and edge node 1 logical station package	Record device state, ingestion time, local events, and process messages.	Select the pilot broker, assign a Pi, and configure authentication and ACLs.

Printed fixtures assigned to this work area

Fixture	Qty	Envelope (mm)	Material / use	Fit gate
WA-4 scale-compatible lid receiving tray	1	240 × 180 × 18	PETG. Defines the receiving and camera region while distributing load on the scale.	Usable region is 232 x 172 mm. Verify the scale platform and largest lid envelope.
WA-4 barcode-scanner stand	1	130 × 100 × 130	PETG. Keeps a handheld scanner in a repeatable position beside the receiving tray.	Handle slot is 40 mm wide. Confirm scanner handle width and trigger clearance.
WA-4 low-voltage decision-button console	1	150 × 80 × 28	PETG. Houses three low-voltage buttons for accept, replace, and manual-review decisions.	Cutouts are 22.5 mm diameter. This enclosure is prohibited for mains voltage or safety circuits.

Receiving controls are process inputs only. Printed button housings are extra-low-voltage interfaces and are prohibited for safety or mains circuits.

Hardware assignment & procurement

The action status is intentionally conservative: inventory-list evidence does not establish identity, availability, installation, configuration, or operation. 25 station lines: 4 acquire, 10 verify, 1 verified-existing, and 10 assign/configure.

Area	Action	Qty	Hardware	Purpose	Evidence / next gate
WA-1	VERIFY	1 unit	P2S 3D printer and local telemetry	Print the enclosure base and provide authoritative job lifecycle data.	Identify the physical unit and test its interface and operational state.
WA-1	ACQUIRE	1 unit	Calibrated plug-level energy meter	Measure printer power and cumulative energy for each print job.	Specify, procure and accept against the documented station requirement.
WA-1	VERIFY	1 unit	MLX90640 thermal array	Observe printer heating and cooling without replacing machine interlocks.	Identify the physical unit and test its interface and operational state.
WA-1	VERIFY	1 unit	XIAO ESP32S3 Sense or IMX335 workpiece camera	Observe base presence and removal in a workpiece-only camera region.	Identify the physical unit and test its interface and operational state.
WA-1	VERIFY	1 monitored slot	Keyestudio Hall Magnetic tool-presence sensor and Nano input	Observe whether the designated process-tool slot is present, absent, or unknown without controlling any machine or safety function.	Identify the physical unit and test its interface and operational state.
WA-1	CONFIGURE	1 station interface	Inspection button or user interface	Record base-quality, release, and reprint decisions explicitly.	Implement and test the adapter or interface, including failure behavior.
WA-1	ASSIGN	1 package	Raspberry Pi edge-node package	Host printer and sensor adapters, timestamps, MQTT, and local buffering.	Allocate one identified unit; record power, network and station.
WA-2	VERIFY	1 system	PAROL6 robot arm and controller	Prepare the electronics kit and expose robot-job lifecycle data.	Identify the physical unit and test its interface and operational state.
WA-2	CONFIGURE	1 set	Instrumented component-bin load-cell set	Detect expected component removals and prepared-kit completion.	Implement and test the adapter or interface, including failure behavior.
WA-2	VERIFY	1 unit	IMX335 or XIAO workpiece camera	Observe component-bin, pickup, and prepared-kit presence.	Identify the physical unit and test its interface and operational state.
WA-2	ACQUIRE	1 system	Independent robot guard and emergency-stop system	Provide an independent safety boundary for powered robot motion.	Specify, procure and accept against the documented station requirement.
WA-2	CONFIGURE	1 adapter	PAROL6 controller event adapter	Bind controller lifecycle and fault records to the robot-job identity.	Implement and test the adapter or interface, including failure behavior.
WA-2	ASSIGN	1 package	Raspberry Pi edge-node package	Host controller and sensor adapters, synchronized time, and local buffering.	Allocate one identified unit; record power, network and station.
WA-3	CONFIGURE	1 mat	Thin-film pressure mat	Observe assembly start and component placement transitions.	Implement and test the adapter or interface, including failure behavior.
WA-3	VERIFY	1 unit	Workpiece camera	Observe assembly state while excluding faces and adjacent stations.	Identify the physical unit and test its interface and operational state.
WA-3	CONFIGURE	1 jig	Arduino Nano serial test jig	Execute the versioned sensor-node test and calibration procedure.	Implement and test the adapter or interface, including failure behavior.
WA-3	VERIFY	1 set	Reference scale and calibration references	Record completed-node weight and support versioned calibration checks.	Identify the physical unit and test its interface and operational state.
WA-3	CONFIGURE	1 station interface	Approval button or user interface	Record explicit deployment approval after a passing test record.	Implement and test the adapter or interface, including failure behavior.
WA-3	ASSIGN	1 package	Raspberry Pi edge-node package	Host the serial receiver, evidence fusion, timestamps, and local buffer.	Allocate one identified unit; record power, network and station.
WA-4	VERIFIED EXISTING	1 system	Management PC	Operate run registration, scheduling, delivery, and outcome logging.	Confirm location, identity, condition and approved station role.
WA-4	ACQUIRE	1 adapter	Read-only inverter and BMS adapter	Provide fresh, unit-qualified photovoltaic and battery snapshots.	Specify, procure and accept against the documented station requirement.
WA-4	VERIFY	1 unit	Network scale	Observe lid receipt and removal after tare.	Identify the physical unit and test its interface and operational state.
WA-4	VERIFY	1 unit	XIAO workpiece camera	Observe lid presence and size cues in the receiving region.	Identify the physical unit and test its interface and operational state.
WA-4	ACQUIRE	1 unit	Two-dimensional identifier scanner	Capture shipment, lid, size, and design-revision identities.	Specify, procure and accept against the documented station requirement.
WA-4	CONFIGURE	1 logical station package	MQTT broker and edge node	Record device state, ingestion time, local events, and process messages.	Implement and test the adapter or interface, including failure behavior.

Purchase first: a calibrated printer energy meter, an independently specified robot guard and emergency-stop system, a read-only inverter/BMS adapter, and a two-dimensional identifier scanner. A second scale is also required if one dedicated WA-3 reference scale cannot be physically verified.

3D-printed non-safety fixtures

16 parametric designs, 46 planned prints. Every design is delivered as FCStd / STEP / STL / 3MF; the manifest and per-part README records are the manufacturing handoff.

Area	Part / file stem	Qty	Envelope (mm)	Material	Purpose / fit assumption
WA-1	WA-1 cooled-part inspection tray wa1_inspection_tray	1	220 × 160 × 16	PETG	Keeps the printed base inside a repeatable camera and load-cell inspection region. Usable inspection region is 212 x 152 mm. Verify the produced base envelope before printing.
WA-1	WA-1 sensor-ready scraper and caliper rack wa1_tool_rack	1	180 × 70 × 68	PETG	Separates hand tools from the cooled-part inspection surface and holds one removable Hall-sensor module behind the designated monitored slot. Seven 10 mm open tool slots are provided. Tool handles wider than 10 mm rest in the front channel. The integrated cassette monitors one designated tool slot only. The official module envelope is 30 x 20 mm; PCB thickness, Hall-element position, connector keep-out, and component height require a physical fit check. Validate magnet distance and every allowed resting orientation; an empty slot is not evidence of correct tool use or operator identity. The presence sensor is research telemetry, not a safety interlock, and must not control the printer or protective functions. Monitoring all seven slots requires seven sensors and verified additional digital-input capacity; the workbook lists five complete sensor kits.
WA-1	WA-1 optical and thermal sensor bridge bracket wa1_sensor_bridge_bracket	1	90 × 55 × 71	PETG	Provides a non-safety adapter surface for a camera and MLX90640 sensor pod. Optical opening is 44 x 28 mm. Hole spacing is a prototype and must be checked against the selected sensor carrier.
WA-2	WA-2 instrumented component bin wa2_component_bin	4	110 × 85 × 55	PETG	Holds one identified component type over a dedicated weighing point. Internal envelope is approximately 104 x 79 x 52 mm. Do not use for loose conductive parts without an ESD-safe liner.
WA-2	WA-2 component-bin locator plate wa2_bin_locator	4	122 × 97 × 8	PETG	Returns each component bin to the same load-cell and camera position. Recess accepts a 110 x 85 mm bin base. Verify screw positions against the load-cell platform before drilling.
WA-2	WA-2 prepared-kit tray wa2_kit_tray	1	220 × 150 × 18	PETG with removable ESD-safe liner	Separates prepared electronics into three observable kit compartments. Three equal nominal compartments are provided. The printed polymer is not assumed to be electrically dissipative.
WA-3	WA-3 base, lid, and kit staging tray wa3_component_staging_tray	1	220 × 150 × 16	Validated ESD-safe PETG or PETG outside the ESD mat	Provides three separated identity-preserving staging pockets before assembly. Each pocket is approximately 65 x 138 mm. Verify component envelopes and surface-resistivity requirements.
WA-3	WA-3 Nano serial test-jig enclosure wa3_nano_test_jig_enclosure	1	110 × 80 × 28	Validated ESD-safe PETG	Protects a low-voltage carrier board and provides controlled USB cable exit. Prototype post pattern is 58 x 36 mm. Carrier-board dimensions and electrical clearances require a physical fit check.
WA-3	WA-3 reference-weight caddy wa3_reference_weight_caddy	1	180 × 70 × 18	PETG	Keeps four calibration references identified and separated from active test parts. Blind pockets have radii 18, 15, 12, and 9 mm. Measure the certified references before production printing.
WA-4	WA-4 scale-compatible lid receiving tray wa4_receiving_tray	1	240 × 180 × 18	PETG	Defines the receiving and camera region while distributing load on the scale. Usable region is 232 x 172 mm. Verify the scale platform and largest lid envelope.
WA-4	WA-4 barcode-scanner stand wa4_barcode_scanner_stand	1	130 × 100 × 130	PETG	Keeps a handheld scanner in a repeatable position beside the receiving tray. Handle slot is 40 mm wide. Confirm scanner handle width and trigger clearance.
WA-4	WA-4 low-voltage decision-button console wa4_button_console	1	150 × 80 × 28	PETG	Houses three low-voltage buttons for accept, replace, and manual-review decisions. Cutouts are 22.5 mm diameter. This enclosure is prohibited for mains voltage or safety circuits.
Shared	Shared workpiece-camera privacy hood shared_camera_privacy_hood	4	76 × 80 × 58	Matte black PETG	Limits glare and narrows incidental visibility outside each workpiece region. Clear optical channel is 68 x 50 mm. The hood supplements but does not replace software privacy masking.
Shared	Shared Raspberry Pi 5 under-table sled shared_pi5_under_table_sled	4	130 × 100 × 18	PETG	Provides ventilated, removable edge-node retention below each work surface. Board-hole pattern is 58 x 49 mm. Verify the selected Pi case, PoE hardware, and NVMe carrier envelope.
Shared	Shared table-edge cable clip shared_table_edge_cable_clip	12	35 × 42 × 36	PETG	Routes extra-low-voltage sensor cables without placing loose loops on work surfaces. Open throat is 25 mm high and 33 mm deep. Use only after measuring the actual table edge; never route mains or safety wiring.
Shared	Shared work-order label-card holder shared_label_card_holder	8	100 × 30 × 42	PETG	Keeps removable work-order and component identity cards visible at transfer points. Card channel is approximately 17 mm deep. Use pseudonymous identifiers when privacy policy requires them.

Use the native files correctly

FCStd is the editable FreeCAD parametric source; **STEP** is the neutral solid exchange; **STL** and **3MF** are mesh exports. Confirm units in millimetres, slicer profile, material, wall count, orientation and supports before release.

Production release gate

Print one fit-check article, measure critical interfaces, record material batch and file revision, inspect for cracking or warpage, then authorize the required quantity. Replace rather than repair a damaged fixture unless a reviewed repair method exists.

All printed parts are non-safety fixtures for extra-low-voltage or passive workholding only. They must not guard a robot, carry a robot, implement an emergency stop, enclose mains voltage, or replace certified protection.

Process orchestration & Munich boundary

The physical route follows the distributed Bayreuth branch in the source BPMN. Machine and safety control remain local; only defined messages cross sites.

Bayreuth distributed sequence

Step	Work area	Mode	Process stage	Physical implication
1	WA-4	sequential	Run And Energy Control	Register the run and derive a schedule from a fresh energy state.
2	WA-4	alternative	Configuration Selection	Select large or standard, or wait for the next solar window and evaluate a fresh energy state.
3	WA-4	sequential	Lid Commitment	Request and evaluate Munich's commitment before production.
4	WA-1	parallel	Base Production And Quality	Produce and accept the base, including review and energy-authorized reprint loops.
4	WA-4	parallel	Remote Lid Receiving	Receive and accept the Munich lid or request a replacement.
5	WA-1 + WA-4	sequential	Parts Ready	Synchronize the accepted base and accepted lid.
6	WA-2	sequential	Kit Preparation	Prepare the electronics kit after both enclosure parts are ready.
7	WA-3	sequential	Assembly Test And Approval	Assemble, test, calibrate, rework when required, and approve.
8	WA-4	alternative	Outcome Control	Apply disclosure for a shared run and complete the process.

Cross-site message contracts

Contract	Direction	Message	Required content
1. lid request	TinyHouse (Bayreuth) → Munich Lab	Scheduled lid specification	work-order ID · energy-schedule ID · size · design revision · due window
2. commitment decision	Munich Lab → TinyHouse (Bayreuth)	Commitment decision	accepted or rejected commitment
3. lid dispatch	Munich Lab → TinyHouse (Bayreuth)	Lid dispatch	shipment ID · component ID · size · design revision
4. delivery decision	TinyHouse (Bayreuth) → Munich Lab	Delivery decision	accepted or replacement-required decision
5. minimized outcome	TinyHouse (Bayreuth) → Munich Lab	Minimized outcome	privacy-minimized completion measures · privacy-minimized sustainability measures

Boundary rules

- Machine and safety control remain local to each laboratory.
- Raw energy telemetry remains in Bayreuth by default.
- Munich receives the selected configuration and due window, not battery history.

Control invariant

- Raw video remains local by default.
- The collaboration coordinator never sends direct printer or robot commands.

BLOCKING SOURCE-MODEL GAP — Task_ProduceLocally has no outgoing sequence flow in Process_TinyHouse. The local topology and rejected-commitment fallback do not reach parts ready, assembly, approval, or Run complete in the source BPMN. No replacement flow is invented in this concept.

Commissioning gates & unresolved evidence

Treat the concept as a controlled design input. Release each station only after its physical, safety, utility, data and privacy evidence is recorded.

Release gates

Gate	Release condition	Required record
G1 · Measured survey	Room, openings, fixed services, tables and operating envelopes measured and photographed	Updated dimensioned plan; deviation register
G2 · Furniture & utilities	Table load/height/brakes, outlet circuits, network ports, ventilation and service access accepted	Signed furniture and utility checklist
G3 · Machinery safety	Printer operating conditions reviewed; PAROL6 risk assessment, certified guarding, interlocks, E-stop and mount validated	Independent safety acceptance; test records
G4 · Sensor & evidence	Each sensor identified, calibrated, timestamped, disconnected/fault-tested and bound to the correct work order	Versioned adapter and calibration records
G5 · Privacy & cybersecurity	Camera regions, retention, ACLs, broker authentication, data minimization and cross-site disclosure approved	Privacy review; network and access-control record
G6 · Dry run	No-power physical route followed; then controlled end-to-end pilot with failure and recovery cases	Pilot evidence package and issue closure

Known source conflicts and explicit handling

Subject	Conflicting evidence	Concept handling
Installed dome-camera identity	The workbook lists one Annke I91DG dome camera. [docs/EDV TinyHouse.xlsx, Tabelle1!A40:C40] The photographed installed dome camera is branded TP-Link VIGI. [docs/public/images/tinyhouse/exterior-side.jpg, visual inspection]	Treat these as potentially different devices. Record model, serial number, network address, and location before assigning either camera.
Installed router identity	The workbook lists one WaveShare industrial 4G LTE router. [docs/EDV TinyHouse.xlsx, Tabelle1!A20:C20] The infrastructure documentation identifies the installed router as TP-Link Archer MR600 V3.0. [docs/network/router-and-switch.md, lines 5-17]	Use the TP-Link documentation for the current installed-router concept. Classify the WaveShare unit as listed until its physical role is verified.
Pineboards HatDrive quantity	The detailed HatDrive row lists quantity ten. [docs/EDV TinyHouse.xlsx, Tabelle1!A29:C29] A later abbreviated HatDrive row also lists quantity ten. [docs/EDV TinyHouse.xlsx, Tabelle1!A44:C44]	Treat the two rows as a possible duplicate. Do not report or assign twenty units until purchase records and physical stock are reconciled.
Rear short-wall window width	The scanned floorplan indicates 160 cm; the existing OBJ geometry uses 150 cm.	Show the 160 cm source-plan value as nominal and require a measured survey before installation.
Table-to-work-area mapping	The table drawing provides four footprints but no work-area assignment.	Label the selected mapping as a design inference and verify heights, loads and condition before move-in.

Primary design-hold-point references

Primary source	URL	Design use and limit
Bambu Lab P2S official specification sheet	Open official source	Physical size 392 × 406 × 478 mm. The 50 × 50 cm plan envelope is a concept fit-test assumption, not a manufacturer-certified installation clearance; verify access, ventilation, door/top, spool and service needs.
Source Robotics PAROL6 official documentation	Open official source	Documents 400 mm reach. The plan reach circle is geometric context only; guarding and mounting remain independent risk-assessment hold points.
Raspberry Pi 5 official mechanical drawing	Open official source	Controls the bare-board geometry and mounting-hole fit check. Verify the selected case, PoE hardware and NVMe carrier before releasing a sled.
Keyestudio KS0487 Hall Magnetic module documentation	Open official source	Specifies digital on/off output, up-to-3 cm magnetic detection and a 30 × 20 mm module envelope. The WA-1 cassette remains provisional until the actual PCB, Hall element, connector and magnet distance are measured.
BAuA ASR A2.3	Open official source	Provides the 0.90 m concept target used for the stated small-occupancy main escape-route check. This report does not establish compliance.
DGUV robot safeguarding guidance	Open official source	Supports the requirement for a risk assessment, protective devices and interlocking. It does not define a guard geometry for this cell.

Definition of a commissioned station

Physical acceptance Physical: labeled table and equipment; fixed access routes clear; rear bay has no tables; operating/service envelopes marked; cables restrained; fire and first-aid equipment unobstructed.	Evidence acceptance Digital: stable device identity; synchronized time; documented units; calibration; work-order binding; disconnect/fault behavior; local buffering; access control and privacy-minimized cross-site messages.
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Completion of this report proves only that a coherent concept artifact was generated from the repository sources. It does not prove dimensions on site, hardware availability, live integrations, safe robot operation, electrical capacity, fire suitability, privacy approval, or production readiness.